

Sajini Tharushika

Information Technology Undergraduate

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LinkedIn: [LinkedIn](#) | **GitHub:** [GitHub](#) | **Portfolio:** [Portfolio](#)

Profile

Information Technology undergraduate at the University of Moratuwa with a focus on software engineering and full-stack development. Hands-on experience developing and deploying web applications, backend services, and distributed systems through academic and individual projects. Strong foundation in OOP, data structures and algorithms, databases, software engineering, and testing, with practical experience in REST APIs, authentication, asynchronous processing, real-time communication, AI integration, and deployment. Seeking a software engineering internship to apply and further develop backend, full-stack, and distributed-systems skills.

Education

University of Moratuwa, Sri Lanka

2023 – 2027

B.Sc. (Hons) in Information Technology

Current GPA: 3.4

Sujatha Vidyalyaya, Matara

G.C.E. Advanced Level – Mathematics Stream

Z-Score: 1.87

Skills

- **Programming:** Java, Python, JavaScript, C++
- **Web & Backend:** Spring Boot, Spring Security, React, Next.js, Node.js, PHP, REST APIs, WebSockets, STOMP, HTML, CSS, Tailwind CSS
- **Databases & ORM:** PostgreSQL, MySQL, MongoDB, Redis, SQL, Prisma, JPA/Hibernate
- **Software Engineering:** Object-Oriented Programming, Data Structures & Algorithms, Software Testing, Authentication, Agile/Scrum
- **Distributed Systems & Infrastructure:** RabbitMQ, Docker, Kubernetes, GitHub Actions, CI/CD, Asynchronous Processing, Observability
- **Tools & Development:** Git, GitHub, VS Code, Vercel, Render
- **AI & Data:** Pandas, NumPy, Scikit-learn, Machine Learning Fundamentals, AI API Integration
- **Additional Technologies:** OpenGL, GLSL, Arduino, ESP32, Embedded Systems

Projects

Nexora – AI-Enhanced Project Management System

Group Project

- Developed a role-based, AI-enhanced project management system designed to streamline project management, team collaboration, task management, and client interactions.
- **My Contribution:** Developed the Administrator Dashboard, user registration and role assignment, Manager-side project creation and management, and an AI Chatbox for conversation summarization and AI-summary-to-ticket creation.
- **Tech:** Java, Spring Boot, Spring Security, React, PostgreSQL, Python, FastAPI, Groq API, RabbitMQ, WebSockets/STOMP, JWT, Docker

FlowForge – API Management & Distributed Workflow Platform

Individual Project

- Developed a production-oriented platform combining API management with distributed workflow processing using Spring Boot, RabbitMQ, and WSO2 API Manager.
- Implemented asynchronous worker processing, retries, dead-letter queues, idempotency, authentication/authorization, Redis-based coordination, and observability for reliable distributed job execution.
- **Tech:** Java, Spring Boot, PostgreSQL, WSO2 API Manager, RabbitMQ, Redis, Docker, Kubernetes, Prometheus, Grafana, GitHub Actions

AURA – Full-Stack Fashion E-Commerce Web Application

Individual Project

- Developed a full-stack e-commerce application with product browsing, cart management, checkout, authentication, rewards, and voucher redemption.
- Implemented backend APIs, secure authentication, and a gamified spin-the-wheel rewards system using Next.js and Prisma.
- **Tech:** Next.js 15, React, JavaScript, Tailwind CSS, Prisma, MongoDB, JWT, bcrypt

LaserTunes – Laser-Based Musical Instrument

Group Project

- Developed an interactive musical instrument using 12 laser beams to trigger musical notes, with instrument switching, percussion effects, and loop recording.
- Integrated Arduino, ESP32, sensors, and audio hardware for real-time musical interaction; presented the project at **FIT EXPO 2025**.
- **Tech:** Arduino Mega, ESP32, C++, VS1053B, Piezo Sensors, Embedded Systems

Solar – Computer Graphics Animation Project

Group Project

- Developed a 3D computer graphics animation using rendering, shaders, textures, and visual effects.
- **Tech:** C, C++, GLSL, OpenGL

Fitness First – Gym Website

Individual Project

- Developed a fitness website with trainer information, services, user authentication, and an interactive BMI calculator.
- Implemented backend functionality, database operations, authentication, and asynchronous frontend communication.
- **Tech:** HTML, CSS, JavaScript, PHP, MySQL

Leadership & Extracurricular Activities

- **Editorial Head** – IEEE Computer Society
- **Web & Technology Pillar Member** – MoraSpirit
- **Editorial Board Member** – Leo Club of University of Moratuwa
- **Event Coordinator** – Leo Club of University of Moratuwa
- **Editorial Volunteer** – Rotaract Club

Referees

Mr. Chirath Perera — Senior Software Engineer, Creative Software

Phone: +94 76 576 3993

Ms. Nipuni Premadasa — Lecturer, University of Moratuwa

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